APPROVED

By the Educational and Methodological Council

LLP "Astana IT University"

Protocol №_ dated "__" _____202_

Appendix to the internal regulatory document
"Guidelines for Diploma Projects/Works at LLP 'Astana IT University,'" dated November 28, 2019

Table 1.1 – Criteria for Project Evaluation under the Educational Program

Parameter	Score	Rationale
Creative Ability	25 points	The project author(s) demonstrate(s) creative ability in developing a new scientific approach to problem-solving (25 points). The author(s) use creative skills to apply existing data analysis methods or creatively use new equipment (20 points). Creative abilities are displayed in data interpretation (10-20 points). A project of a purely referential nature earns up to 10 points.
Scientific Thinking	35 points	Assessment includes: clarity of the work's purpose (3 points); well-defined research questions/hypotheses (3 points); relevance of the research (3 points); object and subject of the study (2.5 points); technical solution's superiority over existing ones (2.5 points); results verified in real conditions (2.5 points); clear direction for the next research stage ensuring problem resolution (1 point); user accessibility (2.5 points); quality of the literature review (15 points): over 20 citations - 15 points; 15-20 citations - 10 points; 10-15 citations - 5 points; under 10 citations - 0 points.
Depth of Topic Coverage	10 points	Score depends on specific criteria: completion level of the planned stages (3 points); coverage of the problem scope (1 point); whether conclusions are based on individual or repeated experiments (2 points); depth and clarity of presentation (2 points); practical feasibility of results (2 points).

Parameter	Score	Rationale
Effectiveness of Research Methods	20 points	Projects using advanced equipment or theoretical methods to obtain groundbreaking results score 15-20 points. Routine methods yielding new insights score 5-15 points, and purely referential works earn up to 5 points.
Oral Skills and Presentation Quality	10 points	Assessed based on: clarity of delivery (3 points); comprehensiveness of the project's presentation (2 points); emphasis on key project points (1 point); clear presentation of experimental data (1 point); clear statement of project results (1 point); quality of supporting materials (2 points).

TOTAL: 100 points

1.2 – Criteria for Practical Project Evaluation under the Media Technologies and Digital Journalism Programs

Students in the Media Technologies and Digital Journalism programs may choose a practical media project format for their final work. The media project should address issues presented in the accompanying academic paper. The main written work must be supplemented with a USB flash drive containing the media project. During the defense, the project should be presented in a completed (final draft) form.

Criteria for Media Project Creation:

a) Documentary Film:

- Historical Accuracy and Argumentation:
  The documentary should not only present facts but also analyze and interpret them. A relevant thesis or argument is essential. The topic should be set within a historical context, encompassing intellectual, social, political, and cultural perspectives, demonstrating causes, effects, and links between local themes and broader historical trends. Diverse perspectives should be presented to show the topic's multidimensionality.
- Use of Primary Sources:
  The film should be based on primary sources directly related to the topic, such as:
  - Scientific articles
  - Personal letters, speeches, diaries
  - Newspapers from the relevant period
  - Oral interviews, historical documents, photographs, artifacts

Using these materials will help provide depth to the topic and create a fact-based narrative.

- Conclusions and Relevance:
  The film should conclude with statements on the social significance of the topic, linking it to broader historical trends and human development.

- Format:
  The video should be in MP4 format (MPEG-4 Part 14), compatible with most browsers and video platforms, including YouTube. Video and audio tracks should be compressed separately to maintain high-quality video (HD) while minimizing file size.
- Duration:
  The film’s duration should be between 7 and 15 minutes.
- Script:
  The film must address a relevant social issue, reflecting the findings and analysis of academic research. The script and storyline should be original and appended to the academic paper.
- Archival Use:
  Archival materials may be used but should not exceed 15% of the total film duration.
- Participation of Characters and Experts:
  The film should include at least two main characters and two experts whose comments or interviews support the film's key ideas. Experts should have an academic degree or at least 10 years of professional experience.
- Subtitles:
  If the film is in Russian or Kazakh, English subtitles are required. All subtitles must be grammatically correct and adhere to spelling norms.
- Credits and Logo:
  The beginning of the film should display the project title and the university logo. The end credits should list the entire creative team (scriptwriters, director, operators, editors, and other contributors).

b) Fiction Film:

- Format:
  The video must be in MP4 format (MPEG-4 Part 14), compatible with most browsers and streaming platforms (e.g., YouTube). Video and audio tracks should be compressed separately, ensuring high-definition (HD) video quality while minimizing file size.
- Duration:
  The film should be 7 to 15 minutes in length.
- Script:
  The script must be based on a relevant social issue, incorporating the conclusions and analysis of the accompanying research paper. The storyline and script should be original and follow a three-act structure. The script must be included as an appendix to the academic paper.
- Cast and Locations:
  The film must feature at least three actors and take place across three different locations.
- Subtitles:
  If the film is in Russian or Kazakh, English subtitles are mandatory. All subtitles, including captions, must be grammatically accurate and conform to standard spelling.

- Credits and Logo:
  The beginning of the film should include the project title and the university logo. The end credits should list the full creative team (scriptwriters, director, operators, editors, and other contributors).

c) Animated Film:

- Duration:
  The film should be 3 to 5 minutes long as an independent piece. If the project includes elements of 2D or 3D animation, the total animation time should be at least 3 minutes, with no repetitive loops.
- Script:
  The script should be based on a relevant social issue or derived from the conclusions and analysis of the research paper. Both the script and storyline must be original and included as an appendix to the academic paper.
- Format:
  The video should be in MP4 format (MPEG-4 Part 14), compatible with most browsers and streaming platforms (e.g., YouTube). Video and audio tracks should be compressed separately to maintain high-quality (HD) video with minimal file size.
- Subtitles:
  If the film is in Russian or Kazakh, English subtitles are mandatory. All captions, including subtitles, must be grammatically correct and adhere to spelling norms.
- Credits and Logo:
  The film should begin with the project title and the university logo. The end credits should list the full creative team (scriptwriters, director, operators, editors, and other participants).

d) Comic Book:

- Script:
  The script should be based on a relevant social issue or derived from the conclusions and analysis of the research paper. The storyline and script must be original, and the script itself should be included as an appendix to the academic paper.
- Comic Format:
  The recommended format is 11x17 inches. In centimeters, this is approximately 16.84 x 26.01 cm.
- Image Resolution:
  A resolution of at least 300 dpi is recommended.
- Margins and Borders:
  Ensure recommended margins and borders are followed to avoid cutting off essential elements during printing or viewing.
- Image Quality:
  - Clarity and Sharpness: Images should be clear, without blurring or pixelation.

    - Color Fidelity: Colors should align with the intended style and maintain quality for both print and digital formats. Consider color profiles for printing (CMYK) and screens (RGB).
- Fonts and Text:
    - Readability: Font size and style should be readable, especially for digital formats.
    - Proper Placement: Text should be appropriately placed within speech bubbles, without overlapping with key image elements.
    - Font Licensing: Font usage should adhere to licensing agreements.
- Panel Layout:
    - Sequence and Logic: Panels should be organized logically to ensure easy reading flow, with transitions that do not confuse the reader.
    - Consistent Spacing: Space between panels should be uniform to avoid clutter or excessive empty spaces.
- Technical Integration of Text and Graphics:
    - Balance of Text and Image: Neither text nor graphics should dominate; a balance is essential for comfortable viewing.
    - Positioning of Sound Effects and Captions: These should complement the visuals without obstructing them.
- Final File Quality:
    - File Format: The comic should be presented in PDF format.
    - Device Compatibility: The comic should display properly on various devices, including computers, tablets, and smartphones (for digital versions).
    - Source Files: Provide source files with all materials and a history of project changes.

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e) Game Development:

- Game Concept and Design:
    - Theme and Plot: The game should have a well-defined theme and storyline, fitting the genre and project objectives. The plot should be logical and engaging.
    - Genre Specification: Identify the game genre (e.g., platformer, RPG, shooter, puzzle) and ensure alignment with the genre's mechanics.
    - Game Mechanics: Define the core mechanics that create engaging player interactions, with at least one innovative mechanic.
    - Target Audience: Clearly identify the target audience and ensure the game meets their expectations and skill level.
- Game Design and Player Interaction (UI/UX):

    - User Interface (UI): The interface should be intuitive, aesthetically appealing, and user-friendly, with easily accessible controls.
    - User Experience (UX): Ensure a smooth flow of actions and ease of interaction with the game world, avoiding unnecessary complexity.
    - Accessibility: Consider the needs of diverse users, such as color-blindness or low vision settings.
- Technical Specifications:
    - Gaming Platform: Specify the platforms (PC, mobile, consoles) the game is designed for and ensure optimization.
    - Game Engine: Identify the game engine (Unity, Unreal Engine, Godot, etc.) and ensure its capabilities match project goals.
    - Performance: The game should run smoothly, with minimal loading times, high frame rates, and efficient memory usage.
- Graphics and Animation:
  Use graphics and animation that provide a high-quality visual experience without straining system performance, ensuring coherence with the game's overall style.
- Programming and Architecture:
    - Modular Code Structure: The code should be organized modularly, allowing each component (physics, AI, UI, etc.) to be independent and maintainable.
    - Version Control: Use version control (e.g., Git) to track code changes and support collaborative development.
    - Code Quality: The code should be clean, well-documented, and follow best programming practices (readability, optimization, error checking).
    - Code Testing: Regular testing of all modules is mandatory to prevent critical errors and bugs.
- Audio-Visual Elements:
    - Sound Effects: Audio should align with in-game actions and enhance immersion, with each sound effect logically tied to player actions.
    - Music: Background music should enhance the gaming experience without distracting from gameplay, with seamless track looping.
- AI and NPC Behavior:
    - AI System: For games with NPCs, develop an AI system that responds realistically to player actions.
    - Behavior Algorithms: Define NPC behavior (friend or foe) based on in-game scenarios, ensuring logical behavior fitting gameplay.
- Testing and Debugging:
    - Functional Testing: Conduct thorough testing to detect bugs and inconsistencies, ensuring all mechanics function correctly.

    - Optimization: Optimize the game for the target platform to minimize lags and technical issues.
    - User Testing: Conduct testing with target audience representatives to gather feedback and improve player experience.
- Final Build and Presentation:
    - Project Completion: The game must be fully developed, with all elements, including graphics, sound, and plot, finely tuned.
    - Presentation: Prepare a demo version to present the game to the assessment committee.
    - Source Files: Provide source files with all materials and a history of project changes.

f) Websites and Platforms:

- Design and Interface:
    - Responsive Design:
      The website should be adaptable for viewing on various devices (desktop, tablet, smartphone) using flexible layout techniques.
    - Intuitive User Interface:
      The user interface (UI) should be user-friendly, with straightforward navigation requiring minimal steps to access main content.
    - Color Scheme and Fonts:
      The color scheme should be harmonious and align with the website's theme. Fonts should be easily readable on all devices.
- Technical Requirements:
    - File Format Support:
      The website should support common multimedia formats (images – JPEG, PNG; video – MP4; audio – MP3), allowing users to upload and view content without issues.
    - Load Speed Optimization:
      The website must be optimized for fast loading times, including image compression and resource caching.
- Subtitles and Localization:
    - Multilingual Support:
      The website should support multiple languages, including subtitles for video content and translation of text into main languages (e.g., English, Russian, Kazakh).
    - Quality Localization:
      All text and subtitles must be grammatically correct and adhere to the standards of the respective languages.
- Information:

    - Branding:
      The top of the website should display the company/university logo and the full project title.
    - Development Team:
      The "About Project" or "Contact" section must include a list of all project contributors (designers, developers, content managers, etc.).
- Interactive Platform Functionality:
    - Registration and Authorization:
      The platform should allow users to register, create personal accounts, and store data for further interaction.
    - Interactive Elements:
      The platform should offer interactive features like forums, chats, commenting, likes, and reposts to increase user engagement.
    - User Settings:
      A personal profile should allow users to customize viewing options, notifications, and personalized content.
- Design and Interface:
    - Modular Structure:
      The platform should have a modular architecture for easy addition of new features like quizzes, polls, content libraries, etc.
    - Responsiveness:
      The platform should display correctly on all devices, providing a consistently high-quality user experience on both mobile and desktop.
- Security:
    - Data Encryption:
      All user data must be protected with SSL and other modern encryption methods.
    - Access Rights:
      The platform should provide different levels of access for users, including administrator, moderator, and user permissions.
- Technical Specifications:
    - File Format Support:
      The platform should support uploading and viewing of various file types, including documents, audio, video, images, and archives.
    - Performance Optimization:
      The platform should minimize delays, especially under high loads and large numbers of simultaneous users.
    - Integration with Other Services:
      The platform should support integration with external platforms (e.g., social networks, cloud storage, online learning systems).
- Content:

- Information Relevance:
  The site should contain up-to-date information on the stated topic, using verified sources like scientific articles, research, or official publications.
- Structure and Logic:
  Content should be presented logically, with clearly defined sections and user-friendly navigation. Each section should unfold sequentially, allowing users to navigate easily through the site structure.
- Multimedia Content Support:
  The website should include images, videos, and infographics for better information absorption. All multimedia content in foreign languages should have captions and subtitles.

g) Mobile Applications

- User Interface and Functionality:
  - Interactivity:
    The application should provide a high level of interactivity, allowing users to engage with various elements without needing to reload pages. Real-time data updates should be supported through asynchronous requests (e.g., AJAX, Fetch API, Axios).
  - UI/UX Design:
    The design of the application should be intuitive and user-friendly, applying modern UI/UX approaches (such as Material Design or Flat Design) to enhance user interaction.
  - Responsive Elements:
    The frontend should support responsive elements, providing instant feedback to users. Single Page Application (SPA) frameworks like React, Vue.js, or Angular are recommended.
- Frontend:
  - Frameworks:
    The application should be built on modern frontend frameworks such as React, Vue.js, Angular, or Svelte to ensure optimal loading speed and responsiveness.
  - Component-Based Approach:
    Use a component-based structure for reusable UI components.
  - SSR and SSG:
    For improved performance and SEO, use Server-Side Rendering (SSR) with frameworks like Next.js or Nuxt.js or Static Site Generation (SSG) to accelerate page loading.
  - PWA (Progressive Web Apps):
    The web application should support PWA features, such as offline functionality, push notifications, and installation as a native app on devices using Service Workers.

- Backend:
    - REST API / GraphQL:
      The backend should provide a REST API or use GraphQL to facilitate flexible client-server interactions. Version control for APIs is essential to prevent disruptions during updates.
    - Real-Time Interaction:
      For applications requiring real-time interaction (e.g., chats or notifications), use WebSocket, Socket.IO, or Firebase for immediate data exchange.
    - Authorization and Authentication:
      Implement authorization and authentication systems using OAuth 2.0, JWT (JSON Web Tokens), Firebase Auth, or session-based methods to protect user data.
    - Microservices Architecture:
      For large-scale applications, consider a microservices architecture, dividing the backend into independent services for users, content, payments, and other modules.
- Database:
    - Relational and NoSQL Databases:
      Depending on the type of data and application tasks, use relational databases (e.g., PostgreSQL, MySQL) for structured data or NoSQL databases (e.g., MongoDB) for flexible, unstructured data.
    - ORM or ODM:
      Use ORM (e.g., Sequelize, TypeORM for SQL) or ODM (e.g., Mongoose for MongoDB) to streamline database interactions and safeguard against SQL injection.
    - Caching:
      Implement caching systems (e.g., Redis, Memcached) to speed up data processing and optimize database interaction.
- General Requirements:
    - Security:
      Use SSL/TLS to encrypt all data transmissions. Protect against XSS, CSRF, SQL injections, and other vulnerabilities using techniques like contextual escaping, input validation, and session protection tokens.
    - Error Handling and Monitoring:
      Log all errors and events using tools like ELK Stack (Elasticsearch, Logstash, Kibana) or Sentry for analysis and quick issue resolution.
    - Testing:
      Automate web application testing using unit testing tools (e.g., Jest, Mocha, Chai) and integration/end-to-end testing tools (e.g., Cypress, Selenium, Puppeteer).
    - CI/CD and DevOps:
      Implement automated deployment and testing using CI/CD tools (e.g., Jenkins,

GitLab CI, CircleCI). Each application version should be tested and deployed automatically to accelerate releases.

- Containerization:
  Use Docker and Kubernetes to containerize and manage the application, ensuring consistent environments across development, testing, and production stages.
- Hosting and Cloud Technologies:
  Use cloud platforms (e.g., AWS, Google Cloud, Microsoft Azure) for hosting, with scalable options based on traffic demands.
- Performance Optimization:
  - Lazy Loading, Code Splitting:
    Implement lazy loading and code splitting to improve loading times.
  - Content Delivery Network (CDN):
    Use a CDN to deliver static assets (images, scripts) closer to users, enhancing performance and reducing server load.
- User Feedback and Analytics:
  - Analytical Tools:
    Integrate tools like Google Analytics, Hotjar, or Mixpanel to analyze user behavior and improve user experience.

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h) Platforms Using Artificial Intelligence (AI)

- User-Centered Design:
  - User Orientation:
    The AI solution should be developed with the needs of the intended users in mind, including user involvement in the design process. The solution should target issues relevant to underserved communities or specific demographic groups and be created collaboratively with those whose data will be used or affected by AI outputs.
  - Transparency and Communication:
    AI-driven decision-making processes must be clearly explained to users and stakeholders, including those without technical backgrounds.
- Understanding the Existing Ecosystem:
  - Contextual Relevance:
    The AI solution should consider the social, economic, and technological context of the target region or demographic. The team should research local needs, existing solutions, and potential impacts on the job market.
  - Legal and Ethical Considerations:
    Students must demonstrate an understanding of local laws, human rights principles, and ethical standards, ensuring compliance with ethical and legal protocols such as privacy protection or GDPR requirements.
- Data Orientation:

    - Data Quality and Bias Management:
      The quality, relevance, and accuracy of data used to train the AI system must be confirmed. Data sets should be appropriate for the context, with any inherent biases effectively managed.
    - Evidence-Based Decision Making:
      AI model decisions should be data-backed, producing reliable and valuable results.
- Use of Open Standards, Data, and Innovation:
    - Openness and Collaboration:
      The AI solution should adhere to open-source principles, with data, algorithms, and innovations accessible for external review and reuse.
- Privacy and Security:
    - Data Protection and Privacy:
      The team should ensure user data protection and privacy throughout the AI lifecycle, including anonymization and secure data storage measures.
    - Risk Management:
      The project should account for potential risks and unintended consequences of data misuse.
- Collaboration and Reusability:
    - Collaboration:
      The team should actively engage with other stakeholders, organizations, or experts, sharing knowledge and contributing to the AI ecosystem.
    - Adaptability and Reusability:
      The AI solution should be designed for adaptability or reuse in different contexts, with documentation for future iterations or use by other developers.

Note: The following media projects will not be accepted by the certification committee: instructional videos, podcast episodes, social/advertisement videos, social media videos, music videos, and music albums.

Other Practical Projects:
Other media technology projects not listed in this document should be discussed with the project supervisor.

When choosing the topic for the diploma project/work, the following should be considered:
● The relevance of the topic and its alignment with the current state and prospects for development in the chosen field;
● The interests and needs of the market: enterprises and organizations that will serve as the basis for the work;
● The scientific and practical significance of the chosen topic.

The diploma project/work must:

- Correspond to the main issues of the specialty for which the student is defending their work;
- Be relevant, contain scientific novelty, and have practical significance;
- Be based on current theoretical and technological achievements in science and practice;
- Be carried out using modern research methods;
- Include research (methodological, practical) sections on the main defended positions;
- Be grounded in advanced international experience in the relevant field of knowledge.

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